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Semantic-aware communication and computation co-optimization in digital twin-assisted Internet of Drones networks using federated deep reinforcement learning

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ABSTRACT

This paper introduces a semantic-aware framework for communication and computation in Internet of Drones (IoD)-assisted digital twin (DT) networks within the context of smart agriculture. In our system, the IoD gathers environmental data and conducts onboard semantic analysis to identify task-relevant features before transmission. We propose a joint optimization scheme for offloading and semantic processing that utilizes federated deep reinforcement learning (FDRL) to address communication and computation constraints via context-adaptive semantic communication. The simulation results show that the proposed approach substantially enhances energy efficiency and communication delay, depending upon the configuration, compared to traditional offloading strategies.

1. Introduction

By 2050, the global population is projected to exceed 10 billion [1], creating food demand beyond current yield growth unless productivity and efficiency improve. Traditional agriculture is constrained by climate variability, limited land, and resource inefficiency, motivating precision agriculture driven by real-time sensing and intelligent control. In this context, Internet of Drones Edge Networks (IoD-EN) use UAVs with onboard sensors for crop monitoring, pest detection, and resource allocation, yet they generate high-volume data that can cause congestion, delay, and excessive energy consumption [2].

To address these issues, we integrate Semantic Communication (SemCom) and Digital Twin (DT) technologies. SemCom extracts task-critical attributes into compact semantic representations [3,4], and drone-based semantic encoders adapt fidelity to preserve essential information under bandwidth limits. In parallel, the DT maintains an updated virtual view of the field, drone operations, device capabilities, and semantic metadata [5–7], enabling proactive optimization and DT-aware learning. Building on this foundation, we propose a federated deep reinforcement learning (FDRL) architecture where drones learn locally while a DT controller aggregates model updates to support privacy and scalability. Raw data remains onboard, and only semantic representations and federated updates are transmitted. We explicitly model the onboard semantic encoding cost and constraints, while achieving lower communication latency and energy by reducing transmitted bits. Our framework distinguishes itself from recent DT-driven FL/RL offloading and AoI-aware FRL papers [8–11] by co-optimizing semantic fidelity and offloading under DT synchronization hints such as freshness and divergence. In [12], the authors present momentum-based FRL update mechanisms to improve convergence speed and aggregation stability, but they do not model DT or semantic payload control. FRL for experience sharing between workers are presented in [13]. Similarly, in [14], the authors presented an approach for multi-task oriented UAV crowd sensing under charging budget constraints, but do not address DT-based SemCom. Our paper contribution centers on DT-aware closed-loop semantic fidelity control and offloading co optimization for bandwidth and energy-constrained smart agriculture. The DT is not a visualization component. It acts as a pre-decision that reshapes the control problem.

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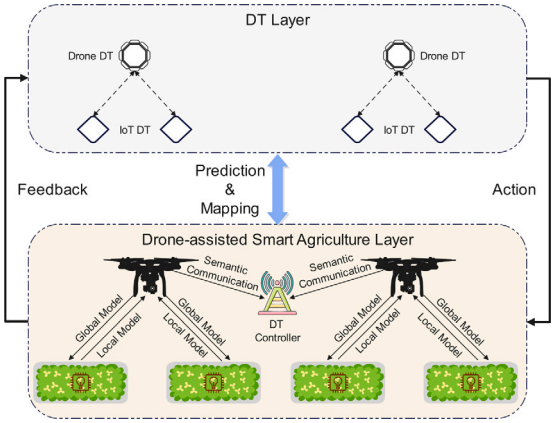

Fig. 1. DT-assisted computation offloading for IoD-EN framework.

It maintains a predicted state from historical and real-time updates, and actions are penalized when the twin is outdated. We explicitly model two DT quality factors. Update latency is the time since the last UAV refresh. Twin divergence is the gap between DT estimates and the physical state. These factors are added in the Markov Decision Process (MDP) state and objective to enable synchronization aware control under uncertainty. The DT-assisted policy anticipates congestion and energy drops and regulates offloading, scheduling, and semantic fidelity via twin freshness. Without the DT, predictive guidance and divergence regularization are absent, and the problem reduces to a standard communication and computation trade off. Building on this formulation, we present a hierarchical FDRL algorithm that jointly optimizes semantic integrity, energy efficiency, and latency by integrating semantic fidelity control with offloading and conditioning decisions on DT freshness and divergence for closed loop resource optimization in UAV assisted edge networks. Communication and computation are tightly coupled in our framework. Semantic fidelity trades off semantic encoding cost against payload size. Offloading shifts execution between local devices and UAV edge servers. Optimizing either component alone can violate bandwidth, CPU and energy constraints. The DT serves as a decision prior that provides predicted state and synchronization quality hints, such as freshness and divergence, which enable robust decisions under mobility. FDRL coordinates learning across distributed UAV regions without centralizing raw data or trajectories by aggregating model updates only. Paper organized as Section 2 presents the system model, Section 3 presents the proposed algorithm, Section 4 presents the setup, and Section 5 presents the results.

2. System model and problem formulation

We consider a smart agriculture IoD-EN with I IoT devices ($i \in \mathcal{I}$) and J drones ($j \in \mathcal{J}$) over slotted time $t \in \mathcal{T}$. Drones act as edge servers and semantic agents, performing local learning and transmitting only compact semantic representations and federated model updates to a centralized DT controller for aggregation and coordination. For FDRL, each drone j collects local experience $\mathcal{D}_j = \{(S(t), A(t), R(t), S(t+1))\}$ from its interactions and associated tasks, and only model updates are aggregated. Multiple drones provide field-wide coverage and distributed edge capacity across spatial IoT clusters, reducing access distance, uplink contention, and per-task latency and energy. Inter-drone cooperation is not assumed, as coordination is realized via DT-guided scheduling and aggregated updates. The proposed framework is shown in Fig. 1.

2.1. Digital twin model

The DT maintains a synchronized virtual replica of the agricultural system, capturing drone mobility, IoT task profiles, semantic fidelity, and resource dynamics. For each IoT device i, the task profile at time t is

$$\text{state}_i^I(t) = \{t_i^{\max}, ar_i(t), size_i(t), env_i(t), c_i(t)\}, \tag{1}$$

where $t_i^{\max}$ is the deadline, $ar_i(t)$ the arrival process, $size_i(t)$ the data size, $env_i(t)$ the context, and $c_i(t)$ the CPU cycles/bit. Each drone j is characterized by

$$\text{state}_j^J(t) = \{loc_j(t), f_j(t), b_j(t), \gamma_j(t), h_j(t)\}, \tag{2}$$

where $loc_j(t)$ is location, $f_j(t)$ CPU frequency, $b_j(t)$ residual energy, $\gamma_j(t)$ semantic fidelity, and $h_j(t)$ channel gain, with stochastic dynamics

$$f_j(t) \sim \mathcal{U}(f_{\min}, f_{\max}), \qquad h_j(t) \sim \mathcal{N}(\mu_h, \sigma_h^2).$$

The DT is refreshed every Δt_t^{sync} seconds and outputs a predictive prior $\hat{\mathbf{x}}_t$ for the next slot. Twin divergence is defined as

$$d_t = \left\|\mathbf{x}_t - \hat{\mathbf{x}}_t\right\|_2, \tag{3}$$

where $\mathbf{x}_t$ stacks realized compact telemetry like mobility, queue summaries, link quality, and semantic context. Both Δt_t^{sync} and d_t improve the MDP state and enter the objective as consistency terms. Notably, $\mathbf{x}_t$ excludes raw sensing data and is piggybacked with semantic payload and model-update messages, enabling d_t estimation without transmitting raw data.

2.2. Semantic encoding and wireless communication models

In each slot t, drone $j \in \mathcal{J}$ acquires raw data $D_j^{\text{raw}}(t)$ over the assigned region. Rather than transmitting full bitstreams, we use a semantic encoder that extracts task-relevant representations and uplinks them over the OFDMA link to the DT controller.

2.2.1. Semantic encoding model

We model the end-to-end semantic encoder as a composition

$$f_{\text{sem}}^{(\gamma)} = \mathcal{E}_{\text{sem}} \circ \mathcal{C}_{\text{sem}}^{(\gamma)} \circ \mathcal{F}_{\text{sem}}. \tag{4}$$

Here $\circ$ denotes function composition, that is, $(\mathcal{A} \circ \mathcal{B})(x) = \mathcal{A}(\mathcal{B}(x))$. $\mathcal{F}_{\text{sem}}$ extracts features, $\mathcal{C}_{\text{sem}}^{(\gamma)}$ performs content-aware compression controlled by fidelity $\gamma \in [0, 1]$, and $\mathcal{E}_{\text{sem}}$ is the entropy coder. We implement $\mathcal{F}_{\text{sem}}$ with a lightweight CNN encoder (MobileNet), suitable for on-board computing. Given a raw input sample x (e.g., an image), $\mathcal{F}_{\text{sem}}(x)$ outputs a feature tensor, $\mathcal{C}_{\text{sem}}^{(\gamma)}(\mathcal{F}_{\text{sem}}(x))$ produces a fidelity-controlled compressed representation, and $\mathcal{E}_{\text{sem}}(\cdot)$ converts it into a transmit bitstream. In implementation, $C(\gamma)$ applies fidelity-controlled compression to the MobileNet bottleneck features before entropy coding. Lower γ increases compression by reducing the retained feature content and representation precision, which directly reduces the transmitted semantic payload size captured by $\rho(\gamma)$ and produces the fidelity-utility trade-off in Fig. 3(a). The semantic representation and the encoded size are

$$S_j(t) = f_{\text{sem}}^{(\gamma_j(t))}\left(D_j^{\text{raw}}(t)\right). \tag{5}$$

$$L_j^{\text{sem}}(t) = \rho_j(t)\,\text{size}\left(D_j^{\text{raw}}(t)\right), \tag{6}$$

Here $\rho_j(t) = \rho(\gamma_j(t)) \in (0, 1]$ is the effective compression ratio and $\rho_j(t) = 1$ means no reduction. Let v_j^{enc} denote the on-board encoding processing rate (bits/s). Then

$$T_j^{\text{enc}}(t) = \frac{L_j^{\text{sem}}(t)}{v_j^{\text{enc}}}, \qquad E_j^{\text{enc}}(t) = \theta_j^{\text{enc}}\, L_j^{\text{sem}}(t), \tag{7}$$

where θ_j^{enc} is the encoding energy per bit. Semantic fidelity affects task utility. We model it by

$$U_j^{\text{sem}}(t) = u\left(\gamma_j(t)\right), \qquad u'(\cdot) > 0,\ u''(\cdot) < 0, \tag{8}$$

In experiments, we set $u(\gamma)$ from validation accuracy versus γ of a MobileNet-based image classification task and normalize it to $[0,1]$, which produces the curves in Fig. 3(a).

2.2.2. Wireless communication model

We consider a standard OFDMA uplink. Let $\mathcal{B}_j(t)$ be the set of resource blocks (RBs) assigned to drone j at slot t, each with bandwidth B_n (Hz). The received SINR on RB n is

$$\mathrm{SINR}_{j,n}(t) = \frac{P_{j,n}(t)\,G_{j,n}(t)}{N_0 B_n + I_{j,n}(t)}, \tag{9}$$

where $P_{j,n}(t)$ is the allocated power, $G_{j,n}(t)$ is the channel power gain including path loss and fading, N_0 is the noise density (W/Hz), and $I_{j,n}(t)$ is the co-channel interference. The per-slot throughput (bits/s) is

$$R_j(t) = \sum_{n\in\mathcal{B}_j(t)} B_n \log_2\big(1+\mathrm{SINR}_{j,n}(t)\big), \tag{10}$$

subject to $\sum_{n\in\mathcal{B}_j(t)} P_{j,n}(t) \le P_j^{\max}$. RBs and powers are assumed to be given by a proportional-fair (PF), while our focus is on semantic payloads and FL. Given $\mathrm{SINR}_{j,n}(t)$ and a chosen modulation–coding scheme (MCS), we obtain a packet-error rate $\mathrm{PER}_j(t)$ from the corresponding lookup curve at the operative packet length. Semantic fidelity $\gamma_j(t)$ affects the payload size $L_j^{\mathrm{sem}}(t)$, but does not updated SINR directly. With stop-and-wait ARQ, the expected number of attempts is

$$A_j(t) = \frac{1}{1-\mathrm{PER}_j(t)}. \tag{11}$$

The goodput-adjusted transmission time for the semantic payload (bits) and the minimal radio-energy model are

$$T_j^{\mathrm{tx}}(t) = A_j(t)\,\frac{L_j^{\mathrm{sem}}(t)}{R_j(t)}. \tag{12}$$

$$E_j^{\mathrm{tx}}(t) = A_j(t)\left(P_j^{\mathrm{tx}}(t)+P_j^{\mathrm{cct}}\right)\frac{L_j^{\mathrm{sem}}(t)}{R_j(t)}. \tag{13}$$

Here $P_j^{\mathrm{tx}}(t)$ and P_j^{cct} is the circuit power. The end-to-end SemCom costs for the drone j are

$$T_j^{\mathrm{semcom}}(t) = T_j^{\mathrm{enc}}(t)+T_j^{\mathrm{tx}}(t), \tag{14}$$

$$E_j^{\mathrm{semcom}}(t) = E_j^{\mathrm{enc}}(t)+E_j^{\mathrm{tx}}(t), \tag{15}$$

In the wireless layer, fidelity effect $L_j^{\mathrm{sem}}(t)$, which then determines delay/energy via (12)–(13).

2.3. Computational and energy model

IoT device i processes a fraction $1-\lambda_i(t)$ locally and offloads $\lambda_i(t)$ to drone j, while drones perform semantic encoding and local learning.

$$\begin{aligned} T_i^{\mathrm{local}}(t) &= \frac{(1-\lambda_i(t))\,\mathrm{size}_i(t)\,c_i(t)}{f_i(t)},\\ E_i^{\mathrm{local}}(t) &= \kappa_i\,(1-\lambda_i(t))\,\mathrm{size}_i(t)\,c_i(t)\,\big(f_i(t)\big)^2, \end{aligned} \tag{16}$$

where κ_i is the effective switching coefficient. Let $D_i^{\mathrm{off}}(t)=\lambda_i(t)\,\mathrm{size}_i(t)$ denote offloaded bits. The uplink delay and energy are

$$\begin{aligned} T_{i\to j}^{\mathrm{ul}}(t) &= \frac{D_i^{\mathrm{off}}(t)}{r_{i\to j}(t)},\\ E_i^{\mathrm{ul}}(t) &= \left(P_i^{\mathrm{tx}}+P_i^{\mathrm{cct}}\right)T_{i\to j}^{\mathrm{ul}}(t), \end{aligned} \tag{17}$$

and the drone execution cost is

$$\begin{aligned} T_j^{\mathrm{proc}}(t) &= \frac{D_i^{\mathrm{off}}(t)\,c_i(t)}{f_j(t)},\\ E_j^{\mathrm{proc}}(t) &= \kappa_j\,D_i^{\mathrm{off}}(t)\,c_i(t)\,\big(f_j(t)\big)^2. \end{aligned} \tag{18}$$

Per-slot edge totals are

$$\begin{aligned} T_{i,j}^{\mathrm{edge}}(t) &= T_{i\to j}^{\mathrm{ul}}(t)+T_j^{\mathrm{proc}}(t)+T_{j\to i}^{\mathrm{dl}}(t),\\ E_{i,j}^{\mathrm{edge}}(t) &= E_i^{\mathrm{ul}}(t)+E_j^{\mathrm{proc}}(t)+E_j^{\mathrm{dl}}(t), \end{aligned} \tag{19}$$

where downlink terms appear only when a same-slot response is required. The network-wide energy is

$$E(t) = \sum_{i\in\mathcal{I}} E_i^{\mathrm{local}}(t) + \sum_{j\in\mathcal{J}}\Big(E_j^{\mathrm{semcom}}(t)+E_j^{\mathrm{train}}(t) + \sum_{i:\,i\to j} E_{i,j}^{\mathrm{edge}}(t)\Big), \tag{20}$$

where $E_j^{\mathrm{semcom}}(t)$ is the drone-side semantic encoding and semantic uplink cost, and $E_j^{\mathrm{train}}(t)$ accounts for on-board learning energy per local update using the same CPU-cycle model that is $E_j^{\mathrm{train}}(t) = \kappa_j\,C_j^{\mathrm{train}}(t)\,\big(f_j(t)\big)^2$ with $C_j^{\mathrm{train}}(t)$ denoting the cycle budget of the local backpropagation step.

2.4. Markov decision process and optimization

To enable dynamic control across slots, we model joint offloading and semantic fidelity selection as an MDP $(\mathcal{S},\mathcal{A},\mathcal{P},\mathcal{R},\beta)$, where $\beta\in(0,1)$.

- **State $\mathcal{S}$.** At time t, the state aggregates compute, energy, channel, and DT consistency

$$S(t) = \left\{\, f_i(t),\, f_j(t),\, b_j(t),\, H_{i\to j}(t),\, Q_i(t),\, \lambda_i(t), \gamma_j(t),\, \hat{\mathbf{x}}_t,\, d_t,\, \Delta t_t^{\mathrm{sync}} \,\right\}. \tag{21}$$

  where f_i, f_j are CPU frequencies, b_j is drone battery, $H_{i\to j}(t)$ summarizes the IoT→drone link, $Q_i(t)$ is device backlog, $\hat{\mathbf{x}}_t$ is the DT prior, d_t is twin divergence, and $\Delta t_t^{\mathrm{sync}}$ is the last DT sync time.
- **Action $\mathcal{A}$.**

$$\mathcal{A}(t) = \Big\{\lambda_i(t),\, \gamma_j(t),\, o_{i,j}(t)\Big\},$$

  where $\lambda_i(t)\in[0,1]$ is the offload ratio, $\gamma_j(t)\in[0,1]$ is semantic fidelity at drone j, and $o_{i,j}(t)\in\{0,1\}$ assigns device i to drone j (if any). Although the action has three components, λ and γ are selected from a small discrete set of levels and $o_{i,j}(t)$ is restricted to a limited candidate UAV set, so the controller selects actions via a single DDQN forward pass over a finite action set rather than exhaustive search. We adopt DDQN since the decision variables are implemented over a finite discrete action set, and DDQN mitigates Q-value overestimation while enabling low-overhead per-slot inference for online offloading and fidelity control.
- **Reward $\mathcal{R}$.** Per-slot reward penalizes latency, energy, and DT inconsistency with $\kappa_d,\kappa_s>0$.

$$R(t) = -\Big(T(t)+E(t)+\kappa_d\,d_t+\kappa_s\,\Delta t_t^{\mathrm{sync}}\Big), \tag{22}$$

- **Transitions $\mathcal{P}$.** Controlled queue, channel, mobility, and DT update dynamics induce $S(t{+}1)$ from $(S(t),A(t))$.

With partitioned execution, the completion time for device i is

$$T_i(t) = \begin{cases} \max\big(T_i^{\mathrm{local}}(t),\, T_{i,j}^{\mathrm{edge}}(t)\big), & \text{if } \exists\, j \text{ with } o_{i,j}(t)=1,\\ T_i^{\mathrm{local}}(t), & \text{otherwise,} \end{cases} \tag{23}$$

and $T(t)=\sum_{i\in\mathcal{I}} T_i(t)$.

Objective. We seek a policy π that maximizes the expected discounted reward

$$\max_{\pi}\ \mathbb{E}_{\pi}\left[\sum_{t=0}^{T}\beta^t R(t)\right]. \tag{24}$$

$$\begin{aligned} \text{s.t.}\quad &(\mathrm{C1})\ 0\le\lambda_i(t)\le 1, && \forall i,\\ &(\mathrm{C2})\ \sum_j o_{i,j}(t)\le 1, && \forall i,\\ &(\mathrm{C3})\ \sum_i o_{i,j}(t)\lambda_i(t)\mathrm{size}_i(t)c_i(t)\le f_j(t)\Delta, && \forall j,\\ &(\mathrm{C4})\ \sum_i o_{i,j}(t)\lambda_i(t)\mathrm{size}_i(t)\le \Omega_j(t), && \forall j,\\ &(\mathrm{C5})\ T_i(t)\le t_i^{\max}, && \forall i,\\ &(\mathrm{C6})\ o_{i,j}(t)\in\{0,1\}, && \forall i,j,\\ &(\mathrm{C7})\ d_t\le\tau_d, \quad (\mathrm{C8})\ \Delta t_t^{\mathrm{sync}}\le\tau_s. \end{aligned} \tag{25}$$

3. Sustainable agriculture resource optimization algorithm (SARA)

While traditional DRL methods, such as DDQN, offer a value-function approximation for control, they face significant limitations like inefficiency, convergence instability, and weak adaptability to environmental variability. SARA addresses these limitations by coupling a DT prior with FRL and SemCom to jointly optimize (i) IoT→drone offloading and (ii) drone-side semantic fidelity. Drones learn from on-board data and uplink only semantic payloads to the DT controller, while IoT devices decide whether to offload to nearby drones. The DT signals $\hat{\mathbf{x}}_t$, d_t, and Δt_t^{sync} add model freshness into control.

3.1. Local level

Each IoT device i observes a local state $S_i(t)$. It selects an offloading ratio $\lambda_i(t) \in [0,1]$ and a serving drone j. We use a shaped immediate reward that reflects the device-side contribution to the network objective:

$$R_i(t) = -\big(T_i(t) + E_i^{\text{local}}(t) + E_i^{\text{ul}}(t)\big), \tag{26}$$

where $E_i^{\text{local}}(t)$ is the local CPU energy, and $E_i^{\text{ul}}(t)$ is the device uplink energy in the IoT→drone path. This section aligns with the global reward and encourages latency- and energy-aware offloading.

3.2. Global level

Drone j aggregates offloaded tasks from its associated devices, executes them, and controls semantic fidelity for its own DT uplink. Its state $S_j(t)$ comprises CPU $f_j(t)$, battery $b_j(t)$, IoT→drone uplink budget $\Omega_j(t)$, assigned load $\sum_i o_{i,j}(t)\lambda_i(t)\text{size}_i(t)$, and DT signals $(\hat{\mathbf{x}}_t, d_t, \Delta t_t^{\text{sync}})$. The action includes fidelity $\gamma_j(t)$, association $o_{i,j}(t) \in \{0,1\}$, and compute scaling $f_j(t)$. The per-drone reward totals:

$$R_j(t) = -\left(\sum_{i:\, o_{i,j}(t)=1} \big(T_{i,j}^{\text{edge}}(t) + E_{i,j}^{\text{edge}}(t)\big) + T_j^{\text{semcom}}(t) + E_j^{\text{semcom}}(t) + \kappa_d d_t + \kappa_s \Delta t_t^{\text{sync}}\right). \tag{27}$$

3.3. Federated training and coordination

IoT devices learn local offloading policies $\{\pi_i\}$ using reward (26), while drones learn global policies $\{\pi_j\}$ using reward (27). The DT controller performs federated aggregation by periodically collecting and averaging drone parameter updates and broadcasting the aggregated model, enabling coordinated learning without raw-data exchange.

3.4. SARA variants for comparative evaluation

To evaluate robustness, we consider three SARA variants. *(1) SARA-Base:* $\alpha = 0.001$, batch size 32, and $\gamma_j = 0.5$. *(2) SARA-FastAdapt:* $\alpha = 0.005$ and batch size 16 for faster adaptation with reduced stability. *(3) SARA-Conserve:* enforces $\gamma_j(t) \le 0.4$ to reduce semantic and transmission energy for power-limited IoDs.

4. Results and evaluation

The simulation and all numeric parameters are summarized in Table 1. Each run lasts 300 s (3×10^4 slots) and results are averaged over five seeds $\{11, 23, 37, 41, 53\}$, reported as mean ± 95% CI. Federated training data are generated online as per-drone transition tuples collected under the mobility, channel, and task-arrival models, which generate naturally non-IID partitions across drones due to spatial association. Drones follow racetrack mobility, and the uplink adopts OFDMA with proportional-fair scheduling. Air-to-ground links use a probabilistic LoS/NLoS pathloss model with shadowing and small-scale fading, and the SINR accounts for thermal noise and co-channel interference from simultaneous uplink transmissions. The rate $r_{i\to j}(t)$ is computed from allocated RBs and the selected MCS under this SINR, consistent with the MCS → PER lookup used for ARQ-aware delay and energy scaling. Reliability is modeled via PER(SINR, MCS, pkt len), and stop-and-wait ARQ yields the expected attempts A=1/(1−PER), which scale transmission time and energy via (12)–(13). Finally, the decision overhead is the bounded inference time of the lightweight DDQN policy and is included as a small additive control latency, so Fig. 2 is dominated by end-to-end communication and execution costs.

Fig. 2(a) compares end-to-end delay, where SARA variants reduce delay relative to DDQN through semantic payload reduction and coordinated policy learning, with FastAdapt achieving the lowest delay and Conserve trading slightly higher delay for efficiency. Fig. 2(b) shows energy versus task count, where SARA variants reduce energy relative to DDQN primarily due to lower transmission load from semantic compression, while FastAdapt consumes more energy than Base and Conserve due to more frequent adaptation. Table 2 summarizes a

Algorithm 1 SARA with DT-Guided Federated RL

Initialize local policies $\{\pi_i\}$ and drone policies $\{\pi_j\}$; set DT prior $(\hat{\mathbf{x}}_0, d_0, \Delta t_0^{\text{sync}})$.
Set fixed RL hyperparameters $\alpha, \beta, B, C, \epsilon_{\min}$ and initialize exploration $\epsilon \leftarrow \epsilon_0$ with decay factor δ.
for global rounds $g = 1, 2, \ldots$ **do**
(Parallel across slots in round)
  for each IoT device $i \in \mathcal{I}$ **do**
    observe $S_i(t)$
    pick $\lambda_i(t)$
    compute local reward $R_i(t)$
    update π_i locally
  end for
  for each drone $j \in \mathcal{J}$ **do**
    collect assigned offloads $\{\, i : \; o_{i,j}(t) = 1 \,\}$
    choose $\gamma_j(t)$ (and $o_{i,j}(t)$ per constraints)
    compute global reward $R_j(t)$
    update π_j locally using learning rate α and batch size B
    update target network every C steps
    update exploration $\epsilon \leftarrow \max(\epsilon_{\min}, \delta\,\epsilon)$
    produce semantic payload $S_j(t)$ and uplink to DT
    receive updated DT prior $(\hat{\mathbf{x}}_{t+1}, d_{t+1}, \Delta t_{t+1}^{\text{sync}})$
  end for
  (Federated aggregation at DT controller) collect $\{\Delta\pi_j\}$, aggregate, and broadcast updated $\{\pi_j\}$
end forreturn $\{\pi_i\}, \{\pi_j\}$

Table 1
Simulation parameters.

Area	1000 × 1000 (m^2)
Drone count N_{Drone}	10
IoT devices N_{dev}	50
Slot duration T_s	10 (ms)
Drone altitude H	120 (m)
Drone speed v	10 (m/s)
Carrier bandwidth	20 (MHz)
RB bandwidth B_n	180 (kHz)
Noise PSD N_0	−174 (dBm/Hz)
Max Drone power $P^{\max}$	23 (dBm)
Scheduler	Proportional-fair (PF)
Path loss & fading	A2G UMi + Rayleigh
Encoder throughput v^{enc}	120 (Mbit/s)
Encode energy/bit θ^{enc}	0.3 (nJ/bit)
Compression ratio $\rho_j(t)$	$\rho_j \in \{0.25, 0.5, 0.75, 1.0\}$
PER model	lookup PER(SINR, MCS, pkt len)
ARQ	stop-and-wait, $A = 1/(1-\text{PER})$
Seeds	{11,23,37,41,53}

Table 2
Enabled components and qualitative trends (Figs. 2–4).

Scheme	SemCom	FL	DT	Key trends
DDQN	No	No	No	High delay/energy/load; low utility/success; slow conv.
Conserve	Yes	Yes	Yes	Lowest load/energy; moderate delay; lower utility (strict γ).
Base	Yes	Yes	Yes	Balanced delay/energy; stable utility/success; moderate conv.
FastAdapt	Yes	Yes	Yes	Lowest delay; fastest conv.; higher energy (frequent updates).

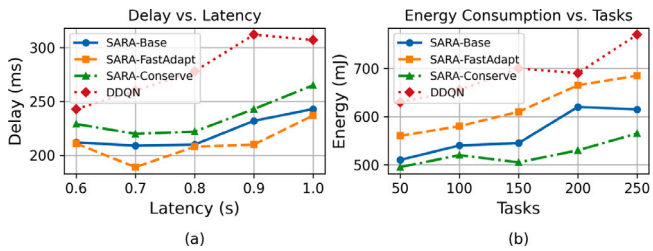

Fig. 2. (a) End-to-End Delay vs Latency and (b) Energy Consumption vs Task Count.

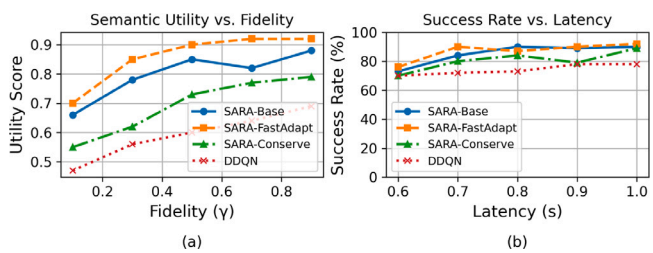

Fig. 3. (a) Semantic Utility vs Fidelity and (b) Task Success Rate vs Latency.

component-wise interpretation of the observed gains in Fig. 2–Fig. 4, which clarifies the roles of SemCom and FL.

Fig. 3(a), semantic utility rises with fidelity across all variations, confirming the efficacy of our semantic encoding. SARA-FastAdapt maximizes semantic value by dynamically adjusting fidelity in response to environmental factors. SARA-Conserve, despite conserving energy, demonstrates reduced semantic utility. DDQN does not attain competitive utility because it lacks semantic encoding. Fig. 3(b) shows an enhancement in task success rates corresponding to increased latency tolerance. SARA-FastAdapt, utilizing adaptive federated learning, attains the best success rate, closely followed by SARA-Base. SARA-Conserve attains somewhat reduced success rates owing to rigorous fidelity restrictions, while DDQN exhibits considerably low performance due to its centralized and non-semantic approach to task management. Fig. 4(a) shows that lower semantic fidelity reduces transmission load by producing smaller semantic payloads, highlighting the utility overhead trade-off. SARA-Conserve yields the lowest load, SARA-Base balances load and utility, and SARA-FastAdapt faces additional encoding and synchronization to maintain utility under dynamics. DDQN performs no semantic compression and therefore faces the highest transmission load. Fig. 4(b) shows the swift convergence of the SARA variations. SARA-FastAdapt achieves convergence swiftly, often within approximately 350 episodes, due to its robust learning configurations. SARA-Base and Conserve achieve stabilization below 600 episodes by utilizing federated aggregation and DT synchronization. In contrast, DDQN requires nearly 1000 episodes, thereby confirming the superiority of our approach compared to traditional centralized DRL techniques. We further evaluate generalization on unseen environments without retraining using $(N_{\text{drone}}, N_{\text{dev}}) \in \{(5,20),(8,40)\}$ and $v \in \{6,8\}$ m/s with bursty arrivals and tighter deadlines. Across these non training environment, SARA preserves the performance ranking and maintains consistent delay and energy gains over DDQN, indicating robustness beyond the training scenario.

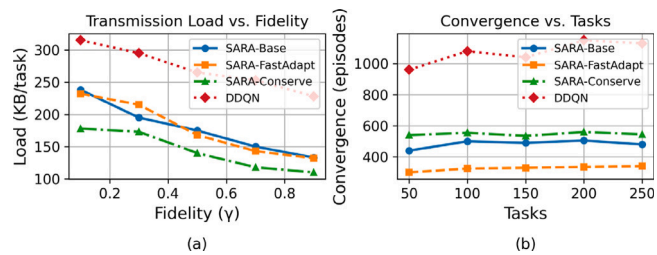

Fig. 4. (a) Transmission Load vs Fidelity and (b) Convergence Episodes vs Task Count.

5. Conclusion

This paper proposes a joint optimization framework that integrates semantic-aware communication into a DT-assisted IoD-EN architecture. The proposed SARA algorithm improves task efficiency and sustainability in smart agriculture through semantic fidelity control and joint offloading. Simulation results demonstrate consistent gains in latency, energy efficiency, and semantic utility over conventional baselines. DDQN is used as a benchmark, however, it does not model semantic fidelity control, DT freshness/divergence, or federated aggregation, which are explicitly optimized in SARA. Future work will extend the evaluation to advanced SemCom-enabled federated RL methods to support a fairer evaluation of generalization.

CRediT authorship contribution statement

Ahmad Arsalan: Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Conceptualization. **Tariq Umer:** Validation, Methodology, Investigation, Conceptualization. **Rana Asif Rehman:** Visualization, Supervision, Project administration, Methodology, Conceptualization. **Byung-Seo Kim:** Supervision, Funding acquisition, Formal analysis. **Mohsin Murtaza:** Funding acquisition, Data curation. **Chi-Tsun Cheng:** Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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